VIVEKANANDA INSTITUTE OF PROFESSIONAL STUDIES

CLASS TEST – MAY - 2023

COURSE: BCA
SUBJECT: Web Based Programming
PAPER CODE: BCA104

SEMESTER: II
M.MARKS: 30
TIME: 1 Hour 30 Minutes

NOTE: - Part – A is compulsory. Attempt any 2 questions from Part B

PART – A 10 Marks

Q. No.	CO	Question	Marks
1.	CO1	Explain Arrays in PHP with suitable example. Differentiate between indexed and associative arrays and write PHP script for displaying the content of 2D array.	(1*10=10)

PART – B 2*10 = 20 Marks

Q. No.	CO	Question	Marks
2	CO2	Differenciate between call by value and call by reference with example. Explain the following functions with suitable examples: preg_match(), preg_match_all(), define(), explode() and compact().	(10 * 1=10)
3	CO1	Differentiate between server-side scripting and client-side scripting languages. What are servers, explain briefly. Differentiate between LAMP, WAMP and MAMP.	(10 * 1=10)
4	CO1	(a) Define different types of scopes of a variable with suitable examples. (b) What are super global variables in PHP? Explain all.	(5 * 2=10)

VIVEKANANDA INSTITUTE OF PROFESSIONAL STUDIES- TC
CLASS TEST - MAY - 2023

COURSE: BCA
SUBJECT: Environmental Studies
PAPER CODE: 110

SEMESTER: II
M.MARKS: 30
TIME: 1 Hour 30 Minutes

NOTE: - Part – A is compulsory. Attempt any 2 questions from part B

PART – A 10 Marks

Q. No.	CO	Question	Marks
1.	CO1	Environment studies is multi-disciplinary by nature. Explain	5
2	CO2	How do various geological activities like Earthquake, Volcanic eruptions etc. occur?	5

PART – B 2*10 = 20 Marks

Q. No.	CO	Question	Marks
2	CO2	What are the major environmental concerns? How can we resolve or reduce them?	10
3	CO3	What is the concept of Sustainability? What are its goals? How can we attain sustainability?	10
4	CO4	Explain environment and its components in detail.	10

...names and job titles of employees who are not working in department 10 and 30

VIVEKANANDA INSTITUTE OF PROFESSIONAL STUDIES-TC
CLASS TEST – MAY - 2023

COURSE: BCA
SUBJECT: Database Management System
PAPER CODE: BCA 108

SEMESTER: II
M.MARKS: 30
TIME: 1 Hour 30 Minutes

NOTE: - Part – A is compulsory. Attempt any 2 questions from part B. Question 3 is divided into part 'a' and part 'b'.

PART – A

10 Marks

Q. No.	CO	Question	Marks
1.	CO 2	Consider the following relational schema: Emp (empno, ename, job, salary, commission, hiredate, deptno) Answer the following SQL queries: a. Display the names of employees working in department 10 and 30. b. Display details of all employees working as MANAGER. c. Display details of all employees with their annual salaries. d. Display details of all employees whose salary range is 10000 to 20000. e. Display the names and job titles of employees who are not working in department 10 and 30.	2*5=10

PART – B

2*10 = 20 Marks

Q. No.	CO	Question	Marks
2	CO 1	Differentiate between the following: i. Generalization and Specialization ii. Cardinality and Ordinality iii. Logical and Physical Data Independence	10
3a	CO 1	Construct an elaborative E-R diagram for a university database considering the constraints given. Take assumptions if required and mention them as footnotes: A university has many departments. Each department has multiple instructors (one person is Head of Department). An instructor belongs to only one department. Each department offers multiple courses, each subject is taught by a single instructor. A student may enroll for many courses offered by different departments.	7
3b	CO 2	Describe the terms: Candidate key, Primary key and Super key.	3
4	CO 1	What are the advantages of using DBMS approach over file system approach. Also, explain the 3-tier architecture.	10

VIVEKANANDA INSTITUTE OF PROFESSIONAL STUDIES - TC
CLASS TEST - MAY - 2023

SEMESTER: II

COURSE: BCA

M.MARKS: 30

SUBJECT: Data Structure and Algorithm using C

TIME: 1 Hour 30 Minutes

PAPER CODE: BCA 106

NOTE: - Part – A is compulsory. Attempt any 2 questions from part B

PART – A

10 Marks

Q. No.	CO	Question	Marks
1. A)	CO1	Suppose an array A [200] [125] is declared as array of integer and base address is 1100. Determine the address of A [100] [75] in row major order.	2
b)	CO2	Evaluate the prefix expression: -,*, 200, 10, 100	2
c)	CO2	Compute the running time of the algorithm for a function that adds two matrices	2
d)	CO2	Convert infix notation into an equivalent postfix notation: P * (Q * R – S) * T – X ↑ Y + (A + C * D)	2
e)	CO1	What do mean by sparse matrix? Explain how can they be represented in memory?	2

PART – B

2*10 = 20 Marks

Q. No.	CO	Question	Marks
Q2 a)	CO2	How is a circular queue advantageous over a standard queue? Explain the insertion and deletion operations in it.	5
b)	CO2	Write an algorithm for the following function to implement a Stack: i) PUSH ii) POP iii) ISEMPTY iv) ISFULL	2 + 2 + 0.5 + 0.5
Q3.	CO2	Define data structures. In how many ways can you categorize data structures? Explain each of them.	10
Q4. a)	CO3	Write an algorithm for Selection Sort and provide an example to verify the algorithm, showing the array after each iteration.	7
b)	CO3	Write a function to search an element in an array using Binary Search.	3

VIVEKANANDA INSTITUTE OF PROFESSIONAL STUDIES- TC

CLASS TEST – MAY – 2023

SEMESTER: Second

M.MARKS: 30

TIME: 1 Hour 30 Minutes

COURSE: BCA

SUBJECT: Applied Mathematics

PAPER CODE: BCA 102

NOTE: - Part – A is compulsory. Attempt any 2 questions from part B. Calculator is allowed. Standard Normal distribution table is allowed.

10 Marks

<table>
<tr><th colspan="3">PART – A
Question</th><th>Marks</th></tr>
<tr><td>Q. No.</td><td>CO</td><td></td><td></td></tr>
<tr><td>1a.</td><td>CO 1</td><td>A and B play a game of dice and the one who throws six will win the game. What is the probability of A's win if he starts the game.</td><td>2</td></tr>
<tr><td>1b.</td><td>CO 1</td><td>The probability that A, B and C solve a particular problem is 0.9, 0.8 and 0.7 respectively. What is the probability that the problem will be solved, if all are independent to each other.</td><td>2</td></tr>
<tr><td>1c.</td><td>CO 1</td><td>The printing error is following the Poisson process. It has been found that there are 4 errors per 10 square feet. 15 square feet is selected at random, find the mean and mode of number of errors.</td><td>2</td></tr>
<tr><td>1d.</td><td>CO 1</td><td>In a lot of 10 items, 2 are defective items and 8 are good quality item. Two items are selected at random. Find the probability distribution of number of defective items.</td><td>2</td></tr>
<tr><td>1e.</td><td>CO 2</td><td>Determine the missing value.<table><tr><td>X</td><td>0</td><td>1</td><td>2</td><td>3</td></tr><tr><td>Y</td><td>10</td><td>-</td><td>18</td><td>22</td></tr></table></td><td>2</td></tr>
</table>

PART – B

2*10 = 20 Marks

<table>
<tr><th>Q. No.</th><th>CO</th><th>Question</th><th>Marks</th></tr>
<tr><td>2 (a)</td><td>CO1</td><td>Mr. A speaks truth 4 out of 5 times. In a game of dice, he reported 6. What is the probability that it is a six?</td><td>5</td></tr>
<tr><td>2(b)</td><td>CO1</td><td>Ms. Y attempt the exam until it is passed. Each time probability of passing an Exam is constant, denoted by P. The probability of failure is Q = 1-P. Find the expected number of attempts.</td><td>5</td></tr>
<tr><td>3(a)</td><td>CO1</td><td>Fit a binomial distribution to the following data<table><tr><td>x</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td></tr><tr><td>f(x)</td><td>28</td><td>62</td><td>46</td><td>10</td><td>4</td></tr></table></td><td>5</td></tr>
<tr><td>3(b)</td><td>CO1</td><td>The average production of 100 one-acre plot is 650 kg with the standard deviation of 25 kg. Find the lowest production of the best 10 plots, if the production is following a normal distribution.</td><td>5</td></tr>
<tr><td>4(a)</td><td>CO2</td><td>Compute the population for the year 2015, the given figures are in millions.<table><tr><td>Year</td><td>1981</td><td>1991</td><td>2001</td><td>2011</td><td>2021</td></tr><tr><td>Population</td><td>68</td><td>79</td><td>92</td><td>105</td><td>130</td></tr></table></td><td>5</td></tr>
<tr><td>4(b)</td><td>CO2</td><td>Fit a 3 degree polynomial that represent the data and hence find f(1.5)<table><tr><td>x</td><td>0</td><td>1</td><td>2</td><td>3</td></tr><tr><td>f(x)</td><td>1</td><td>2</td><td>1</td><td>10</td></tr></table></td><td>5</td></tr>
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